

Week 4 Assignment: Group Policies – Group 6

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Create group policy objects in your lab environment, and write a brief report explaining Kerberos policy settings and group policy processing settings. Submit one report per group.

Part 1: Group Policy

1. Adjust the password policy in the Default Domain policy
 - a. Include a screenshot of the settings

Policy	Policy Setting
 Enforce password history	24 passwords remembered
 Maximum password age	60 days
 Minimum password age	1 days
 Minimum password length	7 characters
 Minimum password length audit	7 characters
 Password must meet complexity requirements	Enabled
 Store passwords using reversible encryption	Disabled

- b. Explain your reasoning for each of the settings you chose.

-Enforce Password History: 24 Passwords Remembered- we chose to leave this setting as default, as when combined with the Maximum password age of 60 days, this makes it so that 24 passwords over the course of 60 days each cannot be reused by users, making it so that for roughly 4 years, those passwords that were previously created cannot be reused. (This is found by multiplying $24 \times 60 = 1,440$ days. Then, $1,440 / 365 = 3.9$ years.)

-Maximum Password Age: 60 days – We do not want passwords to be used indefinitely, and to enhance security we want them to be rotated regularly, but not so often that it is too difficult to keep track of the newly made passwords.

-Minimum Password Age: 1 Day – We do not want users attempting to create multiple new passwords in a short period of time, filling up the 24-password history.

-Minimum Password Length: 7 Characters – We chose to keep this as the standard. Any lower, and it is likely a security risk.

-Minimum Password Length Audit: 7 Characters – We chose to enable this setting, as this will generate a log that keeps track of users that attempt to create passwords that are below our minimum character length.

-Password must meet complexity requirements: Enabled – We chose to keep this enabled as it will enforce the password requirements we have established. It also ensures that passwords are more complex and make use of different characters for higher security.

-Store passwords using reversible encryption: Disabled – We are leaving this disabled as we do not want passwords to be reversible from their encrypted state back into plain text, as this is a massive security risk.

2. Create a folder on the file server
 - a. Create a group policy that will share this folder with a specific department
 - b. Include a screenshot of both your folder and the created policy

ExecutivePlanningDirectory
2/16/2026 4:09 PM File folder

The screenshot shows the Group Policy Management console for the 'Forest: skylogistics.com' domain. The left pane shows the hierarchy: Group Policy Management > Forest: skylogistics.com > Domains > skylogistics.com > Executives > Executives Documents Drive Map. The right pane shows the 'Executives Documents Drive Map' configuration for the 'Executives' group. The 'Scope' tab is selected, showing a list of users and their permissions. Below this, the 'Computer Configuration' and 'User Configuration' sections are expanded, showing the 'Drive Maps' section. The 'Drive Map (Drive: E:)' is configured with the path '\\FILENPRINTSRV\ExecutivePlanningDirectory'.

Link	Order	GPO	Enforced	Link Enabled	GPO Status	WMI Filter	Modified	Domain
	1	Executives Docume...	No	Yes	Enabled	None	2/16/202...	skylogist...

NT AUTHORITY\Authenticated Users	Read (from Security Filtering)	No
NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS	Read	No
NT AUTHORITY\SYSTEM	Edit settings, delete, modify security	No
SKYLOGISTICS\Domain Admins	Edit settings, delete, modify security	No
SKYLOGISTICS\Enterprise Admins	Edit settings, delete, modify security	No

Computer Configuration (Enabled)

No settings defined.

User Configuration (Enabled)

Preferences

Windows Settings

Drive Maps

Drive Map (Drive: E:)

E: (Order: 1)

General

Action	Update
Properties	Update
Letter	E
Location	\\FILENPRINTSRV\ExecutivePlanningDirectory
Reconnect	Disabled
Use first available	Disabled
Hide/Show this drive	No change
Hide/Show all drives	No change

Common

Options

Options	No
Stop processing items on this extension if an error occurs on this item	No
Run in logged-on user's security context (user policy option)	No
Remove this item when it is no longer applied	No
Apply once and do not reapply	No

I signed in from an Executive account to check if the folder is accessible to Executive OU members: (Our Exec OU includes Daniel, Timothy, and Melvin user accounts)

Network locations (1)

ExecutivePlanningDirectory
(\\FILENPRINTSRV) (E:)

The screenshot shows a Command Prompt window with the following text:

```

Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.

C:\Users\daniel>whoami
skylogistics\daniel

C:\Users\daniel>
  
```

3. Create a Password Settings Object

a. Include a screenshot of your PSO.

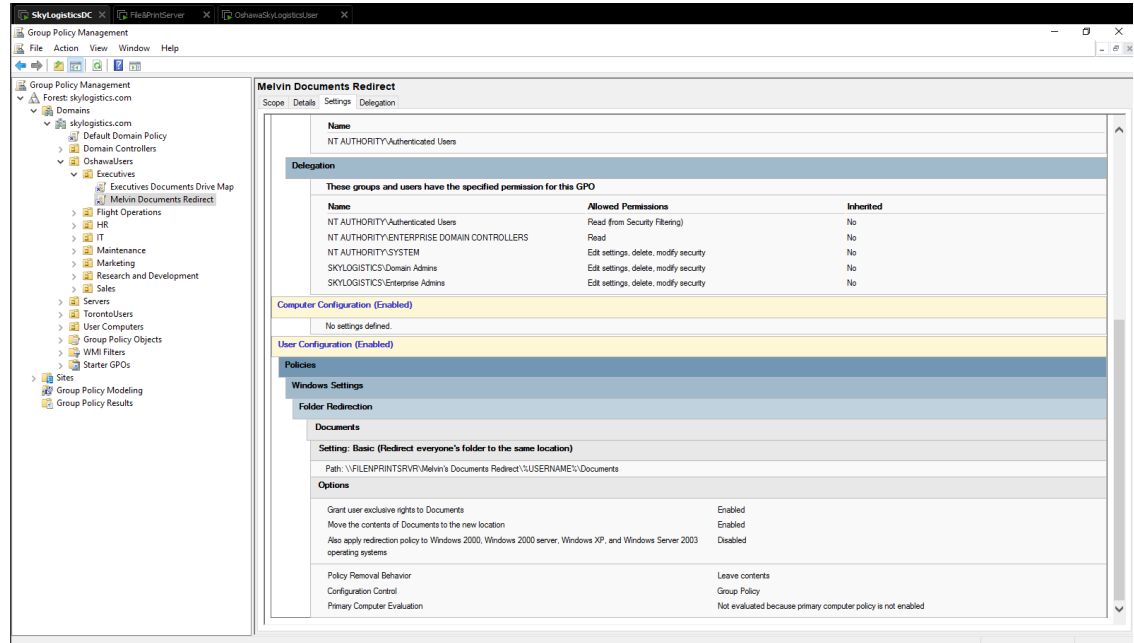
The screenshot shows the 'Create Password Settings: Timothy's PSO' dialog box. The 'Name' is 'Timothy's PSO' and 'Precedence' is '1'. The 'Password age options' section includes: 'Enforce minimum password age' (checked, 5 days), 'Enforce maximum password age' (checked, 90 days), 'Enforce account lockout policy' (unchecked), 'Number of failed logon attempts allowed' (3), 'Reset failed logon attempts count after (m...)' (30), and 'Account will be locked out' (radio button selected for 'For a duration of (mins):' 30). The 'Directly Applies To' section shows a table with one entry: 'Tim'. The 'Description' field is empty. The 'More Information' link is visible at the bottom left. The 'OK' and 'Cancel' buttons are at the bottom right.

Below the dialog box, the Active Directory Administrative Center console is shown. The breadcrumb path is 'Active Directory Administrative Center > skylogistics (local) > System > Password Settings Container'. The console displays a table with one entry: 'Timothy's PSO' with a precedence of '1' and a type of 'Password S...'. The table has columns for 'Name', 'Precedence', 'Type', and 'Description'.

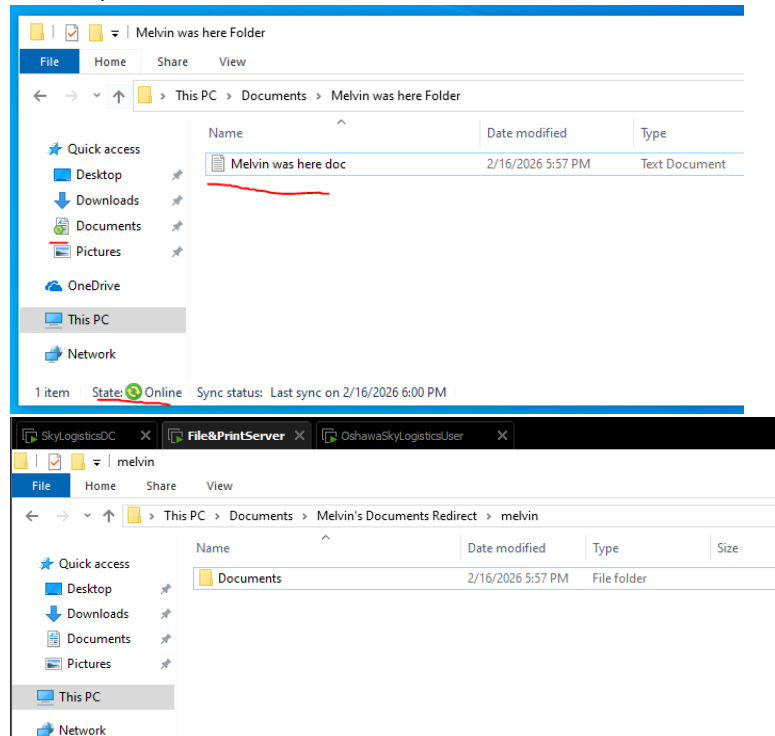
b. How is a password settings object different from a password settings policy?

When it comes to password setting objects, it will supersede the settings of the password settings policy, which is domain wide. The key difference is that PSOs are applied to either users or security groups directly, rather than affecting all users in a domain, including in OU structures like a password policy. PSOs also cannot be applied to OUs. With PSOs you can create customized password settings for specific accounts or groups of accounts.

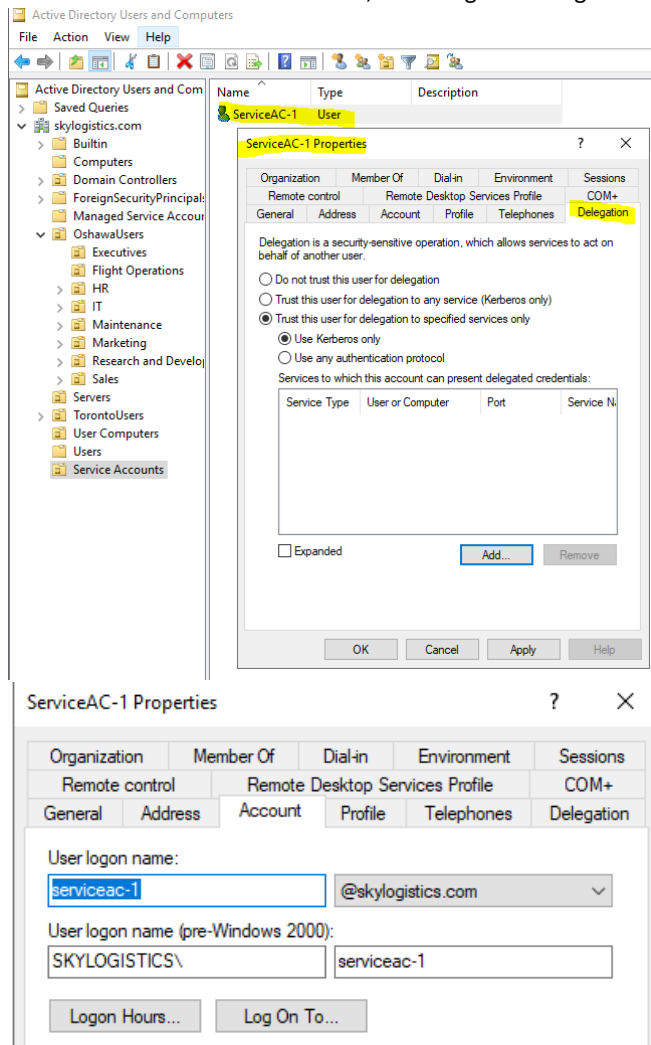
4. Create a policy to redirect the user's Documents folder to a shared folder directory on the file server.
- a. Include a screenshot showing your policy and its settings.



- b. Create a folder on the User's PC and ensure that it is populated in the dedicated File Server directory.



5. Create a user account to be used as a service account
 - a. Assign the user account and SPN.
 - b. Take a screenshot of the account, including the delegation tab.



Part 2: Kerberos

In your own words, write a summary explaining Kerberos. Include:

- a) The components of Kerberos and what each one does

A Kerberos system is made up of what's known as a Key Distribution Center or KDC. The main components of the KDC are the client or user, the user authentication server, the ticket granting server, and the resources that are trying to be accessed.

To access resources on the network, the user interacts first with the authentication server by passing through their login credentials and upon validation provides them with a ticket granting ticket and session key. This TGT is typically encrypted with the user's password and is then given to the ticket granting server. When the user is verified and approved by the KDC, they are then able to access network resources.

- b) Each of the available Kerberos Policy Settings that can be enforced through Group Policy and what they do

There are five Kerberos policy settings that can be enforced through group policy including:

- Enforce user logon restrictions
 - Whether or not the KDC needs to validate every request for a session key against the rights policy of the user
- Maximum lifetime for service ticket
 - How long can a user use a session key before needing to receive a new one
- Maximum lifetime for user ticket
 - How long can a user use a ticket granting ticket before needing to be renewed or receive a new one
- Maximum lifetime for user ticket renewal
 - The amount of time a user can renew a ticket after it has originally been granted
- Maximum tolerance for computer clock synchronization
 - The difference between the user's clock and the domain controller running Kerberos' clock before Kerberos will disallow access

Part 3: Group Policy Processing

1. In your own words, explain:

- a. What the SYSVOL share is on a Domain Controller.

SYSVOL is a shared folder on all domain controllers. This folder holds public files of the domain that include Group Policy Objects, logon and logoff scripts, and other data that must be accessible to clients and all the same across all domain controllers. Any changes made within this folder will be synchronized to all domain controllers so that all users have the same policies or scripts representing the files in the folders.

- b. What is Loopback Processing, and when is it used?

Loopback processing changes the way Group Policy Objects are applied where it is applied based on the location of the computer in the active directory rather than the location of the user. The computer and user configurations would normally apply to the organizational unit of the object during startup and the organizational unit of the user object at logon. Loopback will override the user settings for selected computers.

- c. What is the difference between an Active Directory folder and an Organizational Unit?

An Active Directory folder cannot link or apply Group Policy Objects (GPOs), they cannot grant administrative permissions, have limited nesting, and serves as basic storage. Organizational Units (OUs) can apply GPOs, have admin rights, are fully nestable in other OUs, and have policy control. Administrators replace folders with OUs to apply security settings and policies to computers.

- d. What is slow link processing, and when is it used?

Slow link processing in an Active Directory environment will detect when the bandwidth or latency of a network connection reaches a certain threshold between clients and domain controllers when a Group Policy Object is applied. For example, slow link processing will be used to detect when the bandwidth of the network connection is lower than 1 megabyte per second or 200 millisecond latency. Slow link processing is commonly used with remote users and mobile devices that are susceptible to poor network connections to prevent incomplete applications of policies.

- e. What is the difference between synchronous and asynchronous processing?

Synchronous and asynchronous processing relates to how fast or complete a Group Policy Object is applied when a computer starts or when a user logs on. With synchronous processing, the computer will wait to finish applying policies before the user logon screen appears and the desktop loads. With asynchronous processing, the policies will be applied in the background after the user logs on and the desktop loads. Therefore, synchronous processing has slower logon speeds but more secure while asynchronous processing has faster logon speeds but less secure.